

# WeldSet: An Analytically Constructed Benchmark and Rendered Dataset for Weld-Seam Extraction

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**Abstract**—Welding works along the curve where two parts meet, and a welding robot must find that curve, the seam, in a three-dimensional scan before it can plan a torch path. Published seam-extraction methods report sub-millimetre accuracy, but against hand-taught paths or hand-clicked points, on a few workpieces under one set of conditions, and on data that is rarely shared, so the methods have never been compared on common ground. We present WeldSet, a benchmark in which the ground truth is constructed rather than annotated: scenes of parts waiting to be welded are generated from the seam outward, so every seam is known analytically, every condition (joint type, curvature, groove, thickness, fixture, viewpoint, sensor noise) is a controlled variable, and the same random seed with one variable changed yields an otherwise identical scene. The benchmark corpus has 720 scenes in 12 strata over five joint types and six curved seam families, and a rendered companion set of 36 000 RGB-D frames with seam and tack masks is produced from the same scenes for training learned methods. On the benchmark we reimplement seven published seam-extraction methods and compare them under equal conditions, separating what each method’s geometry can do from what its segmentation front end supplies, and we measure the accuracy of the hand-annotation protocol the field evaluates against. The generator, the corpus, the rendered set, the reimplementations and the evaluation harness are released so that seam-extraction methods, including learned ones, can be compared on the same scenes under the same conditions.

## I. INTRODUCTION

Welding is among the most widely used joining processes in manufacturing, and its automation with industrial robots has moved from mass production, where a taught program is amortised over thousands of identical parts, toward small batches and structural workpieces whose dimensions, deformations and fit-up differ from part to part [1]–[3]. In that setting the robot cannot be taught by hand for every part: it must perceive the workpiece, locate the seam, and plan the torch path and the tack-weld schedule itself [4], [5]. Three-dimensional vision is the sensing modality this literature has converged on: a depth camera on the wrist, a laser line scanner, or a multi-view reconstruction yields a point cloud of the assembly, and the seam is extracted from it as a curve [1], [2], [6]–[8], for regular workpieces and, increasingly, for irregular ones with curved members and intersecting pipes [2], [7], [9].

Deep learning has entered this perception problem more slowly than one would expect from its role elsewhere in robot vision. A few point-cloud networks detect seams directly [10]–[12], and active-perception work chooses where to look [13], but most published pipelines keep the geometry

classical and use a learned network only for a coarse stage: a weld-region segmentation, a per-surface or per-component mask, a bounding box [1], [2], [6], [7], [14]. We see three reasons in how this literature is evaluated, and this paper addresses each.

**The ground truth is not exact.** Published methods are scored against a hand-taught path or hand-clicked points in a scan. One recent method [1] states the error budget of its own truth at about  $0.5\text{ mm}$  and then reports  $0.64\text{ mm}$  RMSE against it. We measured the same annotation protocol with an independent, briefed annotator on scenes whose seams are known exactly and found a median lateral error of  $1.4\text{ mm}$  (Sec. VI). A reported  $0.6\text{ mm}$  against such labels is a statement about the labels.

**Lack of scaling and generalisability.** Each paper evaluates on its own workpieces, with one sensor, one fixture and, with few exceptions, straight seams, so how a method scales beyond them is unknown. There is a lack of work that reports how a method behaves on a curved seam, with the fixture in view, at twice the sensor noise or under a different seed, and the reason is structural: when the truth is tied to a scan, the conditions cannot be varied without re-scanning and re-labelling.

**The data are not available, and would not suffice.** Datasets are typically “available on reasonable request”. Those that are described comprise a handful of workpieces, one acquisition setup and hand-made labels, which is enough to validate a geometric pipeline on its own parts but is neither large enough, varied enough nor accurately enough labelled to train and compare learning-based perception. A method that others cannot run on the same data under the same conditions has not been compared.

Our answer is WeldSet: a scene generator, the benchmark built on it, and one rule that both obey, *ground truth is constructed, never detected*. The generator samples a joint (type, thicknesses, root gap, misalignment, included angle, edge preparation), places the parts by closed-form transforms, and the seam is the intersection of the parts’ surfaces computed algebraically from the same transforms. There is no annotator, no estimator and no “consistency” anywhere in the label path, and a corpus of any size and composition can be generated on demand. Everything else follows from the rule: what is stored, what is a versioned convention, and what is a gate.

The contributions are (i) the benchmark: 720 scenes in 12 strata spanning five joint types and six curved seam families, with per-point visibility and reachability flags, a derived stereo sensor model, paired scenes with and without a clamp-

ing fixture, and gates against dataset shortcuts (Sec. III); (ii) faithful reimplementations of seven published seam-extraction methods, each with the learned front-end stage its paper does not release replaced by an equivalent derived from the truth, so that its geometric mechanism can be measured alone (Sec. IV); (iii) the comparison itself (Sec. V), whose headline is that the dominant mechanism cannot express curvature and that the segmentation stage the papers do not release is most of what they report; (iv) a measurement of the hand-annotation floor of the field’s evaluation protocol (Sec. VI); and (v) the release of the generator, corpus, rendered set, reimplementations and harness as open source, so that others can generate their own corpora, extend the generator, run their own segmentation stages in place of the truth-derived ones, and compare on the rendered tier the same schema produces.

## II. RELATED WORK

**Classical seam extraction from point clouds.** Before learning entered the pipeline, the seam was found by geometry alone: Zhou et al. [4] reconstruct a workpiece model from partial views, denoise it, and extract the seam from the geometry of the reconstruction; Wang et al. [5], [9] recover annular and curved seams from dense reconstructions and line-laser profiles. These pipelines are the ancestors of the coarse-to-fine designs that follow, and the geometric mechanisms below are what survives of them.

**Mechanisms.** We group the seven methods we reimplement by the geometric mechanism that produces the seam, because that is what the benchmark measures. *Plane intersection*: fit planes to the workpiece faces (RANSAC [15] in [1], point-pair-feature voting [16] in [6]) and intersect them pairwise. *Crease detection*: find the points where a local descriptor changes, then chain them; the local bounding-box flatness of [2], [17] and the curvature-seeded region growing of [14]. *Slicing*: cut the cloud into slices, take a geometric centre per slice, fit a spline [7]. *Model registration*: register a CAD-like model to the scan and transfer the seam from the model [3]. *Quadric intersection*: fit quadric surfaces to segmented faces and intersect them [8]. Every one of these pipelines relies on a learned segmentation or detection stage in front of the geometry (PointNet++, FastSAM, K-Net, Faster R-CNN, YOLO or DeepLab in the respective papers), trained on data that is not released; the same coarse-to-fine structure appears in neighbouring tasks such as robotic seam coating [18].

**Learned seam detectors and datasets.** Voxel and point networks detect seams directly [10]–[12], and active-perception work chooses where to look [13]; all are trained and evaluated on scans with hand-made labels and report label *consistency* rather than accuracy. Synthetic data with exact labels is the standard remedy in other perception domains [19], and the value of a shared benchmark with fixed metrics is established [20]; to our knowledge no such benchmark exists for weld seams.

## III. THE BENCHMARK

### A. Constructed truth

A scene is a set of parametric parts (plates, prisms, prepared plates with ISO 9692-1 grooves [21], tubes with planar or saddle cuts, and swept bands along curves) placed by rigid transforms, and its seams are the analytic intersection curves of their surfaces. For plates the seam is the intersection line of two face planes; for a tube on a plate it is an ellipse from the plane–cylinder intersection; for a tube on a tube a saddle curve; for a swept band an offset of its spine. Curved seams enter only from families that admit a two-part realisation: the curve is drawn first and the two parts are derived from it, so the algebraic intersection serves as verification (it rediscovers the constructed curve to  $10^{-13}$ ) rather than as a construction step; grooves are restricted to straight butts, where a swept groove profile cannot self-intersect. That rule, not novelty, is the stopping point of the twelve strata. All seams are evaluated in closed form and stored as dense polylines (0.1 mm spacing) together with the seam tangent, the two face normals, the local dihedral angle and the torch approach direction. Joint parameters follow the standards: thickness and misalignment ranges from ISO 5817 quality levels [22], root gaps scaling with thickness per ISO 9692-1, and the joint taxonomy of ISO 17659 [23], including the distinction between a T joint and an angle joint that most of the literature elides.

Formally, a part  $k$  is a solid whose faces are analytic surfaces  $S_{k,i}$  posed by a rigid transform  $T_k$ , and a seam candidate between parts  $A$  and  $B$  is the curve

$$\gamma = T_A S_{A,i} \cap T_B S_{B,j}, \quad (1)$$

kept if the two faces intersect, face each other across the root gap, or are coplanar and exposed (the third arm exists because coplanar planes never intersect), and if a steerable torch cone (half-angle and standoff) about the bisector of their outward normals is unobstructed; contact between a part and a fixture is stored as a hard negative, never as a seam. For two planes  $\gamma$  is a line, for a plane and a cylinder an ellipse, for two cylinders a saddle curve; each is evaluated in closed form, never fitted. A root gap  $g > 0$  makes “the seam” ambiguous: the face-plane intersection (nominal), the root line on the far face and the midline of the gap are three curves separated by up to  $g$ . We store all three, score against the nominal one, and Sec. VI shows that human annotators click a different one depending on the joint type.

### B. Visibility and the sensor model

A cloud is sampled uniformly by area on every surface, then flagged twice. `visible_from_cam` is geometric visibility from the stored camera: in the frustum, beyond the sensor’s blind zone, facing the camera and unoccluded by an exact ray cast against the parts. `exterior` is reachability by a perfect multi-view scan, from a 30-direction ray scan. Sensor dropout is deliberately not in either flag: a derived stereo noise model (axial  $\sigma_z = sz^2/(fb)$  from a profile’s baseline  $b$ , focal length  $f$  and subpixel precision  $s$ , plus

lateral blur and dropout beyond 70–75° incidence; the D435-class profile is  $b = 50\text{ mm}$ ,  $f = 674\text{ px}$ ,  $s = 0.08\text{ px}$ ) is applied at evaluation time from stored parameters, so that noise is a citable convention rather than a baked artefact and can be scaled as an experimental axis. Three sensor profiles ship; the camera is sampled on a spherical shell whose elevation range deliberately includes views in which a standing plate occludes its own seam.

### C. Rules, gates and twins

Two more labels are *rules* rather than truth, versioned and re-runnable. The first is the *most probable seam* (MPS). In practice a welding robot is first positioned coarsely into a weld region and only then asked to find the seam, so its camera is already inside the approach region rather than sampling an arbitrary viewpoint; that workflow is what makes a single-answer task well posed. The MPS of a view  $v$  is

$$\text{MPS}(v) = \arg \max_{\gamma \in \mathcal{W}} \int_{\gamma} \mathbb{K}[\text{visible}_v(x)] dx, \quad (2)$$

the weldable seam  $\gamma \in \mathcal{W}$  with the greatest visible arclength (ties broken by the fold), evaluated over the exact geometry by a versioned rule. It is a geometric proxy adopted for tractability: which seam carries load is structural knowledge that no single view contains, and we state that scope rather than approximate it. The second rule is a tack-placement schedule (spacing and length as functions of the thinner plate, from shop practice), which yields exact tack positions on every seam. Two corpus-level gates guard against shortcuts: no seam may run parallel to a free plate edge more often than a stated prior allows (outlines are drawn so that the seam-bearing edge is the only axis-aligned one; the corpus passes at a terminal-bin mass of 0.34 against 0.44 allowed), and the tessellated render meshes must stay within 0.25 mm chord error of the analytic surfaces. Every scene has a *twin key*: the hash of the geometry-determining part of its random state, so that a fixture-present twin, a different sensor profile, or a rendered version of the same scene shares bit-identical workpiece geometry and seam truth. The corpus is a pure function of its configuration and seeds; regenerating it under the pinned numerical environment reproduces every content hash. Under a different numpy release the arrays agree to  $10^{-5}$  relative and the hashes do not, which is why the release pins its environment rather than weakening the hash with a tolerance.

### D. Strata, tasks and metrics

The benchmark corpus has 720 scenes, 60 per stratum, in 12 strata: T joints with a line, a pipe (circle), an obliquely cut pipe (ellipse), a pipe on a pipe (saddle), a rectangular tube (rounded rectangle) and a swept stiffener; corner, lap and edge joints; and square, grooved and arc-shaped butt joints (Fig. 1). *Task 1* recovers every weldable seam from a perfect multi-view exterior scan; *Task 2* names the most probable seam from a single view. Predicted polylines  $P$  and true polylines  $G$  are densified to 0.5 mm spacing and matched at

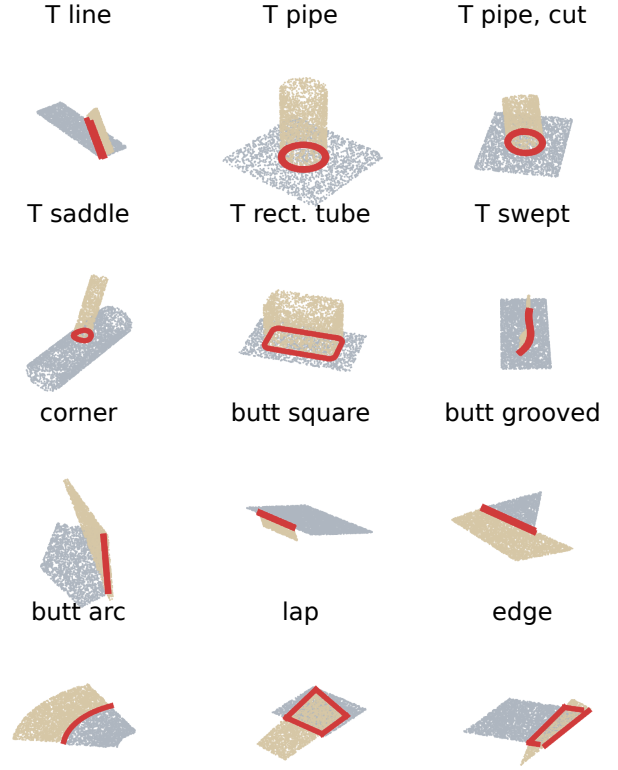


Fig. 1. The twelve strata, one scene each: tier-1 clouds (grey and tan: the two parts) with the constructed seams (red: seams of the joint’s own class; orange: other weldable seams).

a tolerance  $\tau = 3\text{ mm}$ :

$$\begin{aligned} \text{prec} &= |\{p \in P : d(p, G) \leq \tau\}| / |P|, \\ \text{rec} &= |\{g \in G : d(g, P) \leq \tau\}| / |G|, \end{aligned} \quad (3)$$

with F1 their harmonic mean and  $d(\cdot, \cdot)$  the Euclidean distance to the nearest point of the other set; precision penalises phantom seams and recall penalises fragments. We also report the literature’s own matched-path RMSE, the root-mean-square of  $d(p, G)$  over matched points. Every method with a random component is run over 30 seeds per scene and reported as the per-scene median; the deterministic methods show exactly zero spread.

### E. Tiers

Tier 1, used for every result in this paper, is analytic: geometry, ray-cast visibility and the noise model, with no renderer, no simulator and no GPU, because a generator that needs a specific simulator is as inaccessible as the data it criticises. Tier 2 renders the same scenes (RGB, depth, validity, seam, tack and object masks) with published material reflectances, CC0 surface sets and workshop HDRIs, paired to tier 1 by seed and gated so that the rendered clean depth back-projects onto the exact surfaces within 0.25 mm; a 3600-scene training corpus with ten views per scene (36 000 frames) has been rendered and passes the gate on every scene (Fig. 2); it is not evaluated here. The twin

gate has already earned its place: comparing the renderer’s exact depth with tier 1’s visibility flags exposed three errors in our own analytic ray test (thin walls missed by a sampled containment test, a mesh cache keyed before re-posing, exit crossings counted as occluders), each corrected and the corpus regenerated before any number here was computed. Two independent exact implementations of one truth disagree only where one is wrong. Both tiers approach the simulation-to-reality gap from the standards side: ISO joint geometry, a derived sensor model, and a widely supported industrial renderer (Isaac Sim) with published reflectances, so that what is trained on them inherits stated rather than tuned assumptions.

#### IV. SEVEN METHODS, REIMPLEMENTED

Each method was reimplemented from its paper, tested against known geometry, and run through the same harness. Table I lists them with the mechanism and the unpublished coarse stage each depends on. That stage is replaced by an *oracle* from truth of exactly the kind the paper’s network would produce: a weld-region band for M1, per-surface masks for M2, per-component masks for M3, a weld-box crop for M4, per-instance masks for M5, and the welding-surface labels for M7; M6 receives the model itself and is constitutively at the top of the ladder. The oracles are not interchangeable, and supplying the same one to every method would flatter some and starve others; we measured that a seam band has all of its points within 20 mm of a seam while a surface-derived crop of the same scene has 31%, since a plate has six faces and every rim is a surface junction.

Where a paper reports its own numbers, the reimplementation reproduces them: M1 gives 0.60 mm RMSE and 0.81 mm mean error on a T joint at the paper’s point spacing with its weld region supplied, against reported maxima of 0.64 and 1.16 mm, and exactly zero error without noise and with its refit disabled (three exact points give an exact plane, two exact planes an exact line); M3 gives a median 0.51 mm RMSE on corner joints against a reported  $< 0.7$  mm. Two of M1’s equations are unusable as printed and were implemented as the text intends. None of the seven methods is open source, so each was reimplemented from its paper as closely as the text allows, every unstated parameter recorded as such and every open reading kept reachable and priced as a rung (Sec. V). We sought confirmation from the original authors and incorporated the answers we received.

#### V. RESULTS

##### A. Which mechanisms express which joints

Fig. 3(a) is the benchmark’s headline: median F1 per stratum and method under the perfect multi-view scan. Read the rows: a method’s profile across families is its mechanism’s signature.

*The plane-intersection collapse is total where curvature is constant.* M1 holds 0.98 on straight T seams, zero on 4 of the five curved T families and on both curved butt strata, and 0.77 only on the gently swept stiffener, which a plane fits piecewise; M4 degrades more gracefully on curved fillets,

since a ring or a saddle is locally a plane pair for short runs, and dies on butts.

*Three mechanisms survive curvature and one expresses it.* The crease detector (M3), the slicer (M5) and the region grower (M2) all keep a non-zero median on the curved families at a mean of 0.43, 0.72 and 0.67 against the leader’s 0.86; M7, the only quadric surface model, is the best of the six detecting methods on 10 of twelve strata and recovers rings and saddles to 2.53 mm with the paper’s own point selection. Only M6, which is handed the model, exceeds it, and it does so on the strata where registration has the most to grip.

*Grooved butts defeat five of the six detecting methods on the full scan* (median F1  $\leq 0.03$ ); the finding survived rebuilding that stratum with decorrelated outlines, so it is the groove, not the rectangle. The exception is the region grower (0.47), whose crop is drawn around the seam so that the groove walls become the very region boundary it looks for. The ISO 17659 taxonomy exposes one more assumption inside the straight T stratum: between a T joint (3.10, 90°) and an angle joint (3.12, any other included angle) M4 falls from 0.62 to 0.03 at the 15° tolerance we gave its orthogonality test and stays at 0.60 at 30°; the paper states the test but no tolerance, so the number is a property of the gate, which the standard’s distinction reveals and a T-only test set would not. M3 moves from 0.96 to 0.74, M1 from 0.99 to 0.93.

##### B. Reading the profiles through the generator

Because every scene is constructed, each profile in Fig. 3 can be traced to a statement about the method’s model class and a statement about the data. M1 emits a seam only as the intersection line of a near-orthogonal plane pair, so on curved members and on coplanar or parallel butt, lap and edge faces it emits *no candidate at all* (recall zero, not a near miss). M7 is its mirror image: a pipe on a plate is a plane–cylinder intersection and a pipe on a pipe a cylinder–cylinder intersection by construction, so a quadric fitted to samples of a quadric finds the seam wherever the two surfaces meet. Its accuracy is that of its point selection: the paper keeps measured points within a threshold of the fitted intersection (rings and saddles at 1.67–2.53 mm); projecting them onto the fitted intersection, a refinement the paper does not make, would give 0.011 mm (a rung). Where it loses, it loses where one quadric stops describing the face: the rectangular tube is four planes and four fillets (precision 0.51) and the arc butt’s fitted surface over-extends past the seam (0.51). M4’s orthogonal-pair test fires on local tangent-plane pairs all round a cylinder (curved recall  $\geq 0.47$ , precision  $\leq 0.28$ ). M5 takes the geometric centre of each slice along the principal axis of the seam band it is given. Its paper never sees a closed ring whole: each view is an open arc and the arcs are stitched before a closed spline is fitted. Given a whole ring in one band, the axis lies in the ring’s plane, every slice cuts it twice and the centre lands in mid-air (F1  $\leq 0.07$  on the closed families, a rung); given the paper’s arcs it reaches 0.65–0.85, against 0.92 on the open swept seam. M3’s gate is a label test

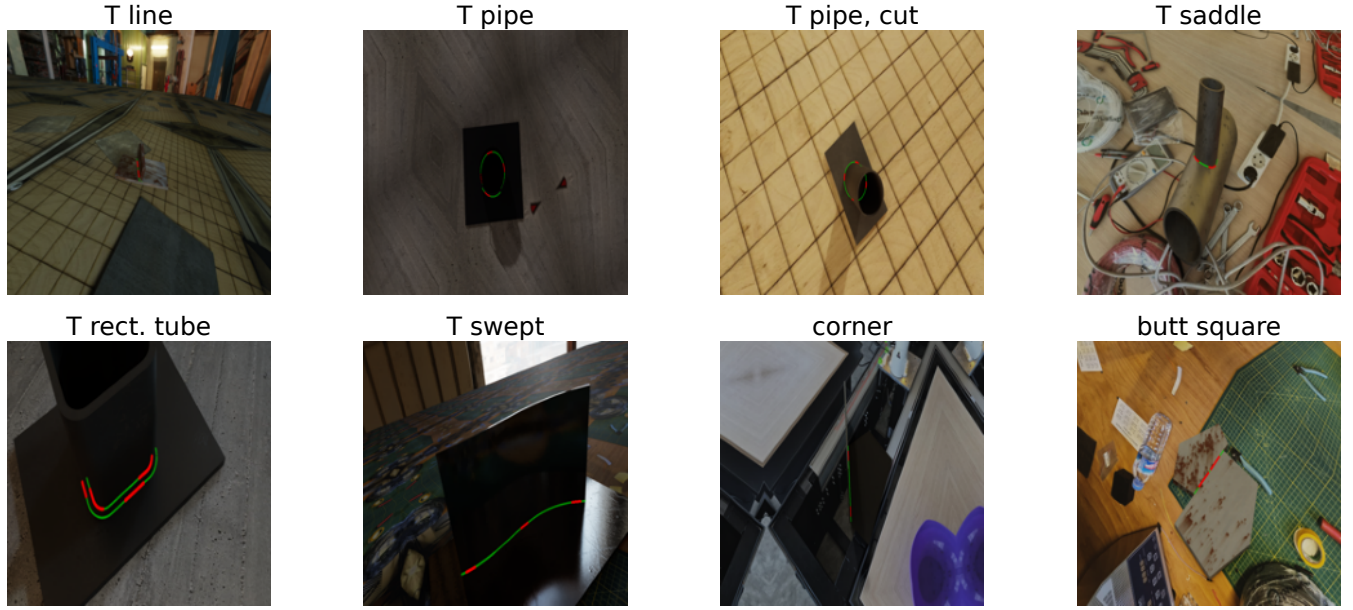


Fig. 2. Tier 2: rendered views of the same benchmark (one per stratum, view 0 = the tier-1 camera), with the constructed seam (green) and tack (red) masks. Not used in the results below.

TABLE I

THE SEVEN METHODS, THEIR MECHANISM, AND THE UNPUBLISHED STAGE EACH DEPENDS ON.

|    | method | mechanism                                          | coarse stage (oracle)    |
|----|--------|----------------------------------------------------|--------------------------|
| M1 | [1]    | multi-plane RANSAC, pairwise intersection          | weld-region segmentation |
| M2 | [14]   | curvature-seeded region growing, edge points       | per-surface masks        |
| M3 | [2]    | bounding-box flatness, K-means, polynomial fit     | per-component masks      |
| M4 | [6]    | PPF coplanarity, orthogonal-pair voting            | weld-box crop            |
| M5 | [7]    | PCA-adaptive slicing, per-slice centres, B-spline  | per-instance masks       |
| M6 | [3]    | CPD registration [24] of a model; seam transferred | the model                |
| M7 | [8]    | quadric fits, surface intersection                 | welding-surface labels   |

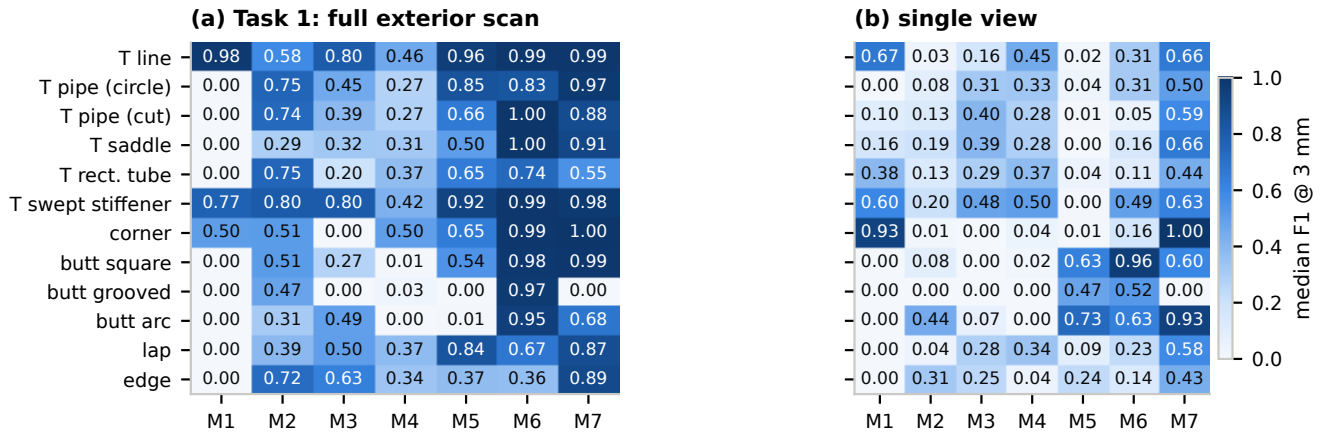


Fig. 3. Median F1 at 3 mm per stratum and method: (a) Task 1, every weldable seam from a perfect exterior scan; (b) the same methods on what one camera returns. M6 is given the model of every scene and has no L1 arm.

(two component labels in one neighbourhood) rather than a geometric one, and the neighbourhood is its paper's  $3 \times 3$  pixel window, about 1.5 point spacings: it cannot see across a root gap wider than that, so corners and butts, whose gaps here reach 4.9 mm, return nothing (0.00 on corners against

1.00 at a 3 mm window, a rung), while edge joints gain (0.63 against 0.22). The paper's corners touch; the window is a joint parameter it never had to state. Closedness costs M3 recall, since its final fit is a single-valued polynomial along the dominant axis. M2's crop is its paper's: prompted by

a weld-keypoint detector, it keeps about 13% of the cloud around the seam. Given a surface crop four times wider, every rim of a six-faced plate becomes a junction of two surfaces and it over-detects everywhere ( $F1 \leq 0.28$ , a rung). With its own crop it scores 0.58 on the straight T and 0.51 on corners and is the one detector that reads the groove (0.47); the price is a coarse stage that carries more than any other in the ladder (0.58  $\rightarrow$  0.11). Finally, the single view is not uniformly harder: M1 rises from 0.50 to 0.93 on corners and M5 from 0.01 to 0.73 on arc butts, because occlusion removes the competing coplanar or parallel face that defeated the method on the full scan; that is a property of coverage, not of robustness, and a benchmark without exact visibility flags could not tell the two apart.

Three properties of the generator flatter every number above and are stated here rather than discovered by a reviewer: surfaces are exact analytic primitives, so a fit to them has no model error and will not survive mill scale or distortion; parts carry no burrs, holes or edge radii, so every precision is an upper bound; and the seam-bearing edge is the only axis-aligned one, which hands the slicing and PCA methods a near-canonical axis. A fourth is the reading: an unstated input or tolerance is priced as rungs, not chosen silently. The benchmark measures what a mechanism *can* express; a real scan measures what it does.

### C. The oracle ladder

Fig. 4 withholds each paper’s coarse stage and changes nothing else. The pooled median F1 of M7 falls from 0.87 to 0.00, of M5 from 0.64 to 0.00, of M2 from 0.58 to 0.11 and of M3 from 0.40 to 0.03. Methods that share nothing but that one input fail by the same factor without it, which makes the honest framing a *ladder every method sits on* rather than a weakness of one. M4 has one more rung: with the generator’s exact normals in place of estimated ones its butt and edge scores rise by about 0.5, so its mechanism is sound and the normal estimate on a thin coplanar pair is what fails. M6’s rungs are its registration choices: with dense features in place of the paper’s sparse ones it slides along the butt overlap (0.97  $\rightarrow$  0.15), and a global initialisation lands on the mirror-image assembly.

### D. Sensor noise

Fig. 5 scales the derived stereo noise on the single view (clean, the profile’s  $\sigma$ , twice it). M7 is essentially noise-immune (0.58  $\rightarrow$  0.50 pooled), M3 does not degrade either (0.22  $\rightarrow$  0.30: the grazing-incidence dropout removes the rim points that feed its phantom creases), and the plane-intersection methods go to zero at the first noise step.

### E. Task 2, the fixture, repeatability and cost

Table II scores the most-probable-seam task: whether the named seam received a prediction, and the matched path error. Two methods sit at 100% selection for a reason that is not skill (M5 receives one mask per seam instance, M6 transfers every seam from the model); their localisation is

the number to read. M7 localises the named seam within 1.5 mm on every class.

The fixture twins (Table III) show where a clamping plate in view lands: not on the plane methods, whose band oracle shields them, but on the crease detector, the region grower and the slicer, which read the fixture-contact lines as seams (M3 adds a median of 22 phantom seams), and on M7, which reads the fixture as a part. Repeatability is measurable here and not reported in the field: a single-draw M1 number is not a measurement, since on 5% of full-scan scenes the same cloud gives F1 anywhere from 0 to above 0.9 depending on the seed alone. Every method runs in well under 4 s per scene.

## VI. THE ANNOTATION FLOOR

Twenty scenes, four per joint type, were exported as clouds and hand-labelled in CloudCompare by an independent annotator who was briefed with the protocol of the annotation-based literature and did not know the truth. Scored against the exact seams (Fig. 6): the median lateral RMSE is 1.4 mm with a 95th percentile of 11.8 mm; endpoint error is worse still (4.2 mm median); 0.20 of the seams were missed and the flush stack of an edge joint invited marking every visible edge. The lap joint is catastrophic for a human as well (8.5 mm median): the plate edge is picked instead of the toe, the same class that every method above struggles with. Which of the three candidate curves the annotator clicked depends on the joint type exactly as the physical crease predicts: on butts the labels are 1.3 mm from the gap midline against 3.4 mm from the nominal seam, on fillets 1.3 mm from the root against 2.0 mm; phantom seams (0.63 per true seam) outnumber misses. The protocol is the field’s own, in CloudCompare because the learned detectors were labelled there, with a 15 mm acceptance gate so that a wide miss counts as a phantom rather than being absorbed, and with the toolchain validated on a synthetic annotator with injected 0.8 mm noise, one planted miss and one planted phantom. The author’s own pass, excluded from the measurement, was worse (1.9 mm), which supports the exclusion rule rather than undermining it. A sub-millimetre accuracy measured against labels with 1.4 mm of noise is below its own label noise.

## VII. DISCUSSION AND LIMITATIONS

**What the comparison says.** Two findings are structural. The plane-intersection mechanism cannot express curvature, and most of the field’s reported accuracy is the accuracy of a segmentation stage that is not released. The improvement with the highest ceiling is therefore not a new geometric primitive but a welding-surface segmentation that feeds a quadric intersection, and the benchmark makes that claim testable rather than rhetorical. The second structural finding is about evaluation: the protocol the field scores against has a 1.4 mm floor, and our exact truth is what makes the floor measurable at all. A third is methodological: several pipelines are underspecified at the stage that decides their score, and the benchmark prices each reading of it as a rung.

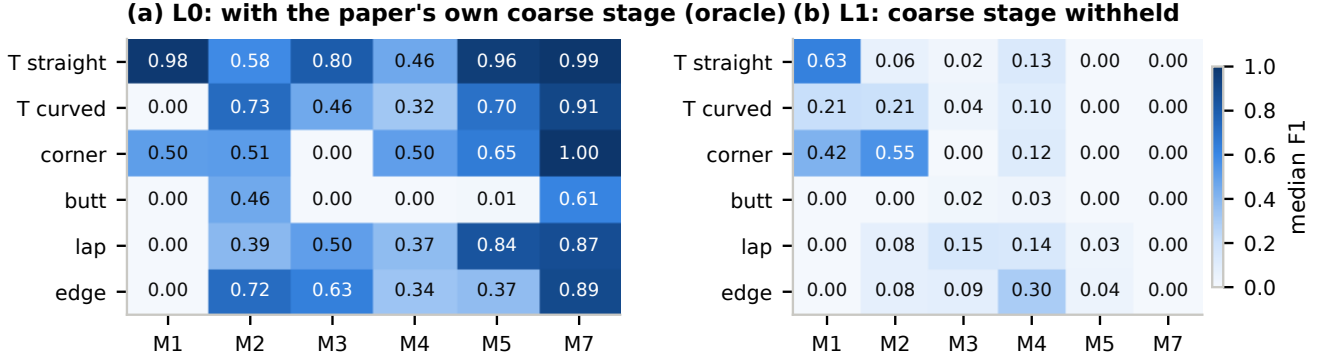


Fig. 4. The oracle ladder: median F1 per joint type (T split into its straight and its five curved families), (a) with each paper’s coarse stage supplied as an oracle and (b) with it withheld. M6 has no withheld arm: its coarse stage is the model, without which there is nothing to register; its registration rungs are in the text.

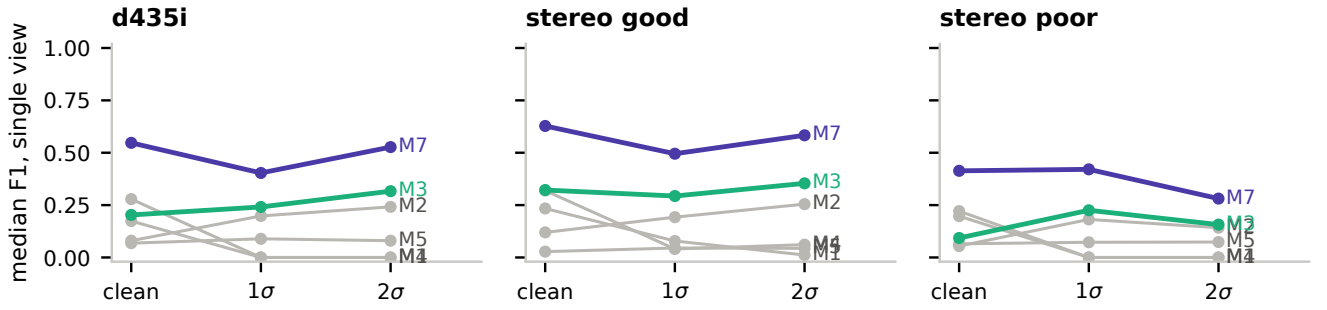


Fig. 5. Median F1 on the single view against the derived stereo noise, per sensor profile. The two leaders are drawn in colour.

TABLE II

TASK 2: SELECTION RATE OF THE MOST PROBABLE SEAM AND MEDIAN RMSE (MM) ON THE MATCHED SEAM, PER JOINT TYPE.

|    | T    | selection rate |      |      |      | T    | RMSE (mm) |      |      |      |  |
|----|------|----------------|------|------|------|------|-----------|------|------|------|--|
|    |      | corner         | butt | lap  | edge |      | corner    | butt | lap  | edge |  |
| M1 | 0.81 | 0.61           | 0.43 | 0.36 | 0.08 | 0.80 | 0.00      | 2.79 | 0.88 | 1.96 |  |
| M2 | 0.72 | 0.57           | 0.76 | 0.78 | 0.83 | 0.49 | 6.22      | 3.05 | 2.44 | 1.27 |  |
| M3 | 0.79 | 0.30           | 0.29 | 0.63 | 0.80 | 0.64 | 0.60      | 0.67 | 0.71 | 1.16 |  |
| M4 | 0.90 | 0.53           | 0.49 | 0.73 | 0.52 | 5.81 | 0.24      | 6.77 | 3.59 | 2.26 |  |
| M5 | 1.00 | 1.00           | 1.00 | 1.00 | 1.00 | 9.73 | 6.80      | 4.13 | 5.31 | 8.72 |  |
| M6 | 1.00 | 1.00           | 1.00 | 1.00 | 1.00 | 4.46 | 4.65      | 1.86 | 4.29 | 6.39 |  |
| M7 | 0.93 | 0.25           | 0.46 | 0.80 | 0.77 | 1.55 | 1.24      | 1.30 | 1.42 | 1.28 |  |

TABLE III

THE FIXTURE PRICE: PAIRED TWINS WITH AND WITHOUT THE CLAMPING PLATE IN VIEW (MEDIAN OVER PAIRS).

|    | pairs | $\Delta F1$ | $\Delta precision$ | extra seams |
|----|-------|-------------|--------------------|-------------|
| M1 | 40    | 0.00        | 0.00               | 0.00        |
| M2 | 40    | -0.14       | -0.23              | 8.50        |
| M3 | 40    | 0.00        | 0.00               | 22.00       |
| M4 | 40    | -0.01       | -0.02              | 0.00        |
| M5 | 40    | -0.20       | -0.18              | -1.00       |
| M6 | 40    | 0.02        | 0.00               | -1.00       |
| M7 | 40    | -0.29       | -0.47              | 4.00        |

**Limitations.** The benchmark is synthetic, and the numbers above are tier-1 numbers. Three answers: the tiered design exists precisely because a physical dataset cannot

vary these conditions, and the rendered tier and a scanned real subset with known part poses are the next two steps of the same schema; the reimplementations are documented readings with tests and, for one method, the first author’s written endorsement, but they are reimplementations; and the most-probable-seam label is a geometric proxy, a stated convention, not a statement about the load-bearing seam. Our own extractor is deliberately not among the seven; the same harness scores it below 0.05 precision on edge joints, a finding harder to dismiss for indicting its authors. The annotation floor is one annotator’s; an intra-rater repeat, separating accuracy from the consistency the literature reports, is still owed.

**What the convention opens.** The findings bound what geometry alone can do; the ceiling they point to is a segmentation stage, and the rendered tier exists so that one

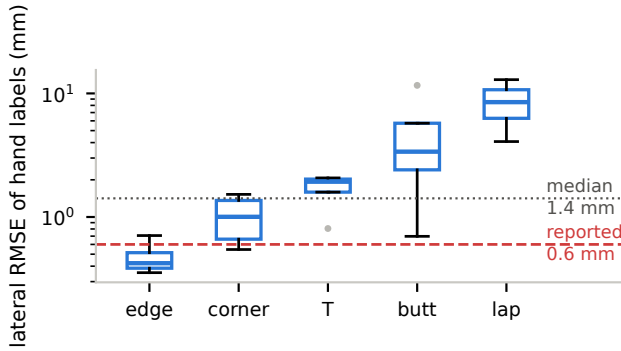


Fig. 6. Lateral RMSE of hand labels against exact seams, per joint type (briefed annotator, 20 scenes), against the accuracy the annotation-based literature reports.

can be trained and then scored, on the same seeds, by the same harness, against the oracle rung it replaces. Three steps follow: an extractor of our own drawing on the mechanisms compared here, a learned detector trained on the rendered tier and scored here, and a real dataset scanned from metal parts with known poses. The generator’s stopping rule admits any seam family two parametric parts can realise, so industrial members (flanges, gussets, stiffener crossings, prepared pipe ends) are extensions of the schema, and the scanned real subset closes the loop to measured truth.

## VIII. CONCLUSION

A benchmark whose truth is constructed rather than detected turns three questions the field could not ask into measurements: which mechanisms express which joints, what the unpublished segmentation stage is worth, and how accurate the labels are that published accuracies are measured against. The generator, corpus, rendered set, reimplementations, harness and results notebook will be released under an open licence through the authors’ institutional repository, mirrored on Zenodo (link withheld for anonymity).

## ACKNOWLEDGMENT

Generative AI (Anthropic Claude, via Claude Code) was used to draft the text of every section from the project’s own documentation and results, to write the figure-generation and evaluation code, and to edit the manuscript; every number, figure and claim was produced by the released code from the released data and checked by the authors.

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